Ingest pit_stops.json file

Step 1 - Read the JSON file using the spark dataframe reader API

```
dbutils.widgets.text("p_data_source", "")
v_data_source = dbutils.widgets.get("p_data_source")
```

```
dbutils.widgets.text("p_file_date", "2021-03-28")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
from pyspark.sql.types import StructType, StructField, IntegerType, StringType
```

```
pit_stops_schema = StructType(fields=[StructField("raceId", IntegerType(), False),
                                      StructField("driverId", IntegerType(), True),
                                      StructField("stop", StringType(), True),
                                      StructField("lap", IntegerType(), True),
                                      StructField("time", StringType(), True),
                                      StructField("duration", StringType(), True),
                                      StructField("milliseconds", IntegerType(), True)
                                     ])
```

```
pit_stops_df = spark.read \
.schema(pit_stops_schema) \
.option("multiLine", True) \
.json(f"{raw_folder_path}/{v_file_date}/pit_stops.json")
```

```
# pit_stops_df = spark.read \
# .option("multiLine", True) \
# .json(f"{raw_folder_path}/pit_stops.json")
```

```
display(pit_stops_df)
```

Table

	raceId	driverId	stop	lap	time	duration	milliseconds
1	1053	839	1	1	15:05:16	30.866	30866
2	1053	20	1	3	15:10:09	32.024	32024
3	1053	854	1	5	15:15:11	51.007	51007
4	1053	853	1	12	15:27:20	31.168	31168
5	1053	842	1	14	15:30:10	31.068	31068
6	1053	20	2	20	15:39:11	31.184	31184
7	1053	854	2	21	15:41:24	32.479	32479
8	1053	20	3	22	15:42:52	39.502	39502
9	1053	853	2	23	15:45:20	31.5	31500
10	1053	852	1	25	15:46:39	30.696	30696
11	1053	847	1	26	15:47:53	29.983	29983
12	1053	8	1	26	15:47:55	30.28	30280
13	1053	830	1	27	15:47:59	29.809	29809
14	1053	842	2	26	15:48:45	32.277	32277
15	1053	832	1	27	15:48:51	30.856	30856

56 rows

Step 2 - Rename columns and add new columns

- Rename driverId and raceId
- Add ingestion_date with current timestamp

```
pit_stops_with_ingestion_date_df = add_ingestion_date(pit_stops_df)
```

```
from pyspark.sql.functions import current_timestamp, lit
```

```
final_df = pit_stops_df.withColumnRenamed("driverId", "driver_id") \
.withColumnRenamed("raceId", "race_id") \
.withColumn("ingestion_date", current_timestamp()) \
.withColumn("data_source", lit(v_data_source)) \
.withColumn("file_date", lit(v_file_date))
```

Step 3 - Write to output to processed container in parquet format

```
# final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_processed.pit_stops")
```

```
# overwrite_partition(final_df, 'f1_processed', 'pit_stops', 'race_id')
```

```
merge_condition = "tgt.race_id = src.race_id AND tgt.driver_id = src.driver_id AND tgt.stop = src.stop AND tgt.race_id = src.race_i
merge_delta_data(final_df, 'f1_processed', 'pit_stops', processed_folder_path, merge_condition, 'race_id')
```

```
display(final_df)
```

Table

	race_id	driver_id	stop	lap	time	duration	milliseconds	ingestion_date
1	1053	839	1	1	15:05:16	30.866	30866	2024-07-24T13:48:32
2	1053	20	1	3	15:10:09	32.024	32024	2024-07-24T13:48:32
3	1053	854	1	5	15:15:11	51.007	51007	2024-07-24T13:48:32
4	1053	853	1	12	15:27:20	31.168	31168	2024-07-24T13:48:32
5	1053	842	1	14	15:30:10	31.068	31068	2024-07-24T13:48:32
6	1053	20	2	20	15:39:11	31.184	31184	2024-07-24T13:48:32
7	1053	854	2	21	15:41:24	32.479	32479	2024-07-24T13:48:32
8	1053	20	3	22	15:42:52	39.502	39502	2024-07-24T13:48:32
9	1053	853	2	23	15:45:20	31.5	31500	2024-07-24T13:48:32
10	1053	852	1	25	15:46:39	30.696	30696	2024-07-24T13:48:32
11	1053	847	1	26	15:47:53	29.983	29983	2024-07-24T13:48:32
12	1053	8	1	26	15:47:55	30.28	30280	2024-07-24T13:48:32
13	1053	830	1	27	15:47:59	29.809	29809	2024-07-24T13:48:32
14	1053	842	2	26	15:48:45	32.277	32277	2024-07-24T13:48:32

56 rows

```
%sql
select * from f1_processed.pit_stops;
```

Table

	race_id	driver_id	stop	lap	time	duration	milliseconds	ingestion_date
1	936	833	1	6	14:15:43	26.392	26392	2024-07-24T13:47:14
2	936	822	1	13	14:25:47	22.974	22974	2024-07-24T13:47:14
3	936	826	1	13	14:25:51	22.072	22072	2024-07-24T13:47:14
4	936	13	1	14	14:27:28	27.936	27936	2024-07-24T13:47:14
5	936	830	1	14	14:27:30	22.536	22536	2024-07-24T13:47:14
6	936	18	1	14	14:27:33	22.231	22231	2024-07-24T13:47:14
7	936	154	1	14	14:27:34	22.805	22805	2024-07-24T13:47:14
8	936	831	1	14	14:27:41	22.729	22729	2024-07-24T13:47:14
9	936	807	1	15	14:28:49	22.573	22573	2024-07-24T13:47:14
10	936	832	1	15	14:28:58	25.096	25096	2024-07-24T13:47:14
11	936	4	1	15	14:29:00	21.567	21567	2024-07-24T13:47:14
12	936	815	1	16	14:30:25	22.068	22068	2024-07-24T13:47:14
13	936	828	1	16	14:30:36	23.005	23005	2024-07-24T13:47:14
14	936	1	1	19	14:34:42	23.146	23146	2024-07-24T13:47:14

8,126 rows

```
Success
```